

Case - Alternative Optimal Solutions

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:53

Objective parallel to a binding constraint: at the optimum a non-basic variable has a zero coefficient in row 0, so another optimal vertex exists.

Model

Maximize $Z = 2X_1 + 4X_2$
subject to
[C1] $X_1 + 2X_2 \leq 5$
[C2] $X_1 + X_2 \leq 4$
 $X_1, X_2 \geq 0$

RESULT

OPTIMAL

$Z^* = 10$

$X_1 = 0$

$X_2 = 2.5$

iterations: 1 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 10$
- Alternative optima: non-basic X_1 has a zero coefficient in row 0 - bringing it in changes the solution but not Z .

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. C1 (Type I, $\leq$): add slack $X3 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X3$ is the starting basic variable of this row.
2. C2 (Type I, $\leq$): add slack $X4 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X4$ is the starting basic variable of this row.

Maximize $Z = 2X1 + 4X2$
subject to
 $X1 + 2X2 + X3 = 5$
 $X1 + X2 + X4 = 4$
 $X1, X2, X3, X4 \geq 0$

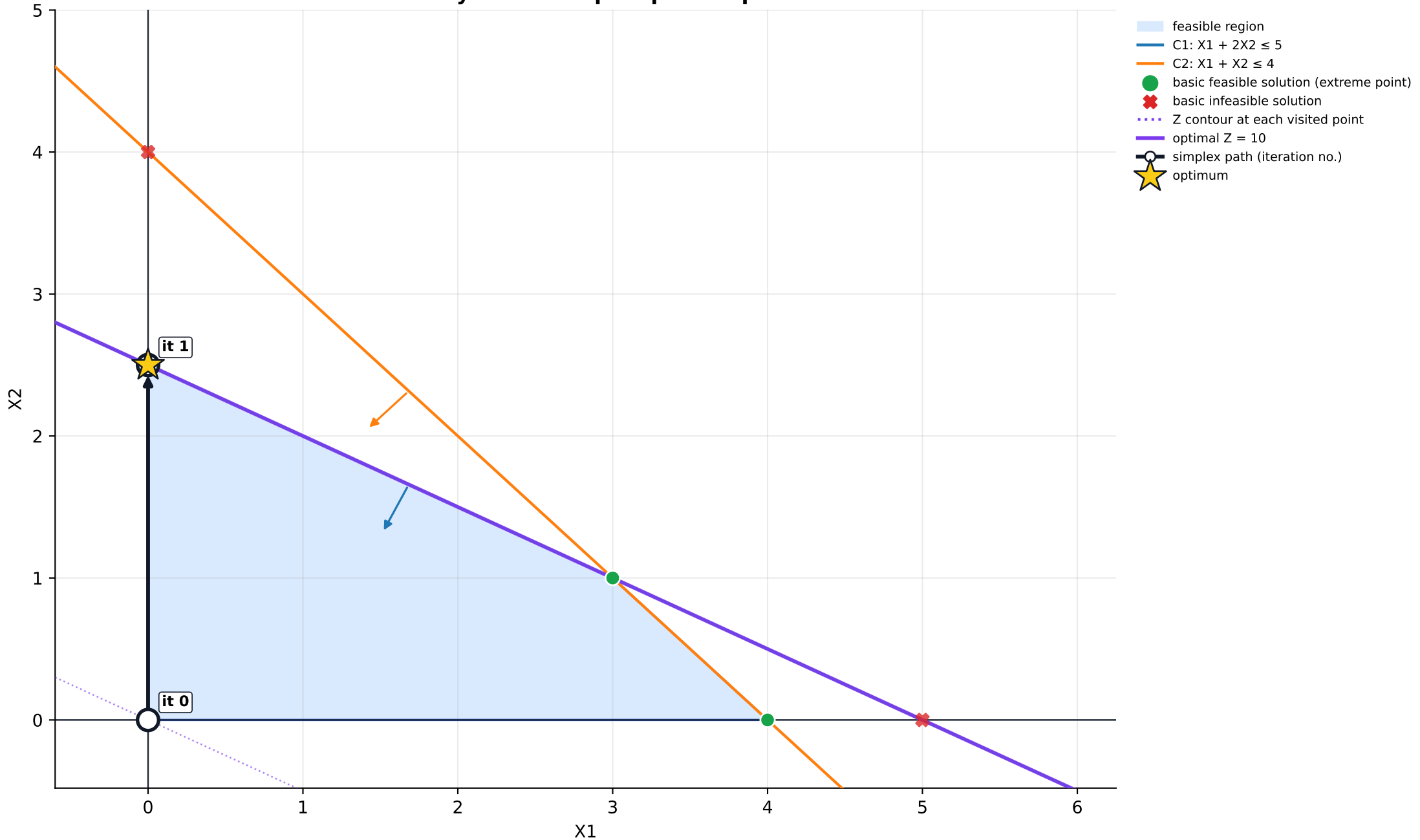
Standard form - coefficient matrix

	X1	X2	X3	X4		RHS
max Z	2	4	0	0		
C1	1	2	1	0	=	5
C2	1	1	0	1	=	4

2. Geometric view: basic solutions, extreme points and the simplex path

improving Z

Geometry of the simplex path - optimal



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	Feasible?	Z	X3	X4	Optimal?
X1, X2	3	1	yes	10	0	0	★
X1, X3	4	0	yes	8	1	0	
X1, X4	5	0	no	10	0	-1	
X2, X3	0	4	no	16	-3	0	
X2, X4	0	2.5	yes	10	0	1.5	★
X3, X4	0	0	yes	0	5	4	

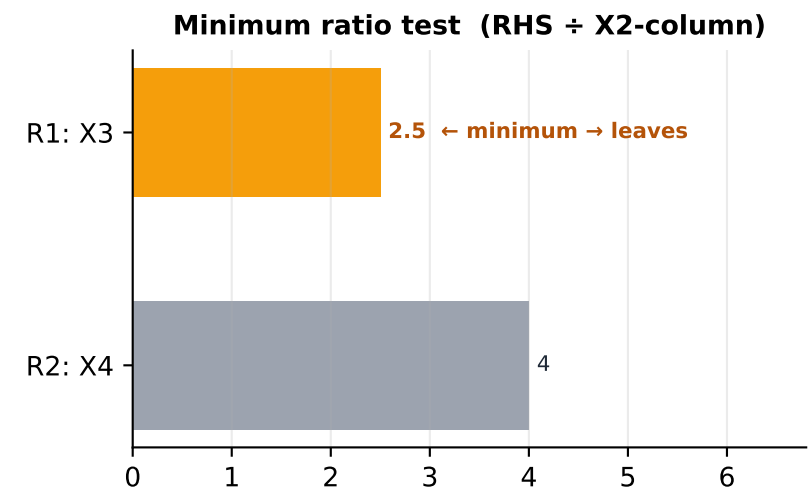
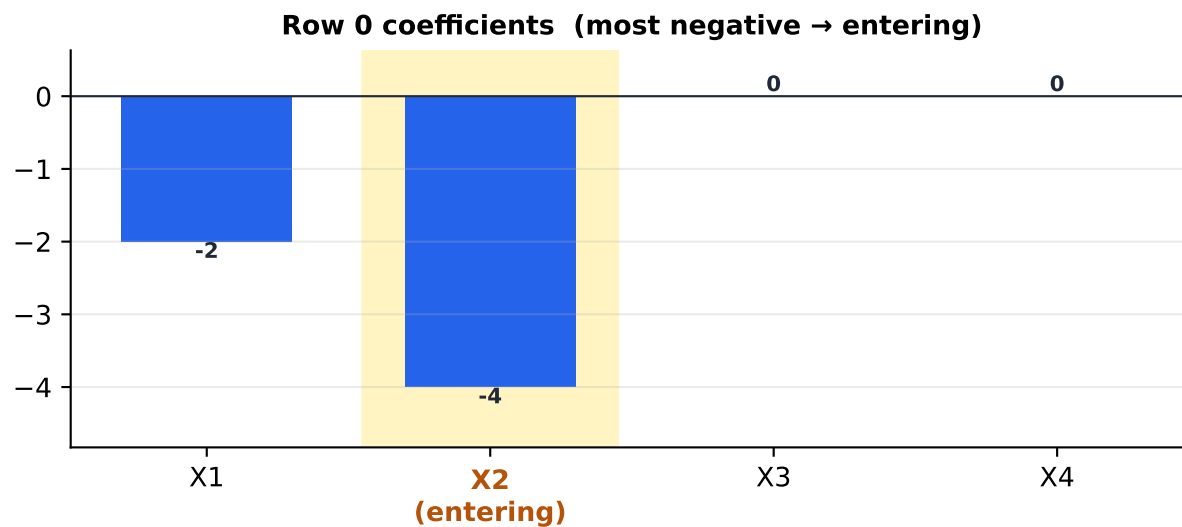
Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	-2	-4	0	0	0
X3	1	0	1	2	1	0	5
X4	2	0	1	1	0	1	4

Initial tableau. Basic variables: $X_3 = 5$, $X_4 = 4$. Non-basic (set to 0): X_1 , X_2 . The values of the basic variables are read directly from the RHS column.

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	-2	-4	0	0	0	
X3	1	0	1	2	1	0	5	2.5
X4	2	0	1	1	0	1	4	4



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 (-4).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: 2.5, R2: 4.
- Leaving basic variable: X3 (row 1, smallest ratio). Pivot number = 2.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R1 by the pivot number 2

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	-2	-4	0	0	0	
X3	1	0	0.5	1	0.5	0	2.5	2.5
X4	2	0	1	1	0	1	4	4

(b) X2 replaced X3 in the basis. Z = 10

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	0	0	2	0	10
X2	1	0	0.5	1	0.5	0	2.5
X4	2	0	0.5	0	-0.5	1	1.5

Row operations:

- Add $4 \times R1$ to R0
- Subtract $1 \times R1$ from R2

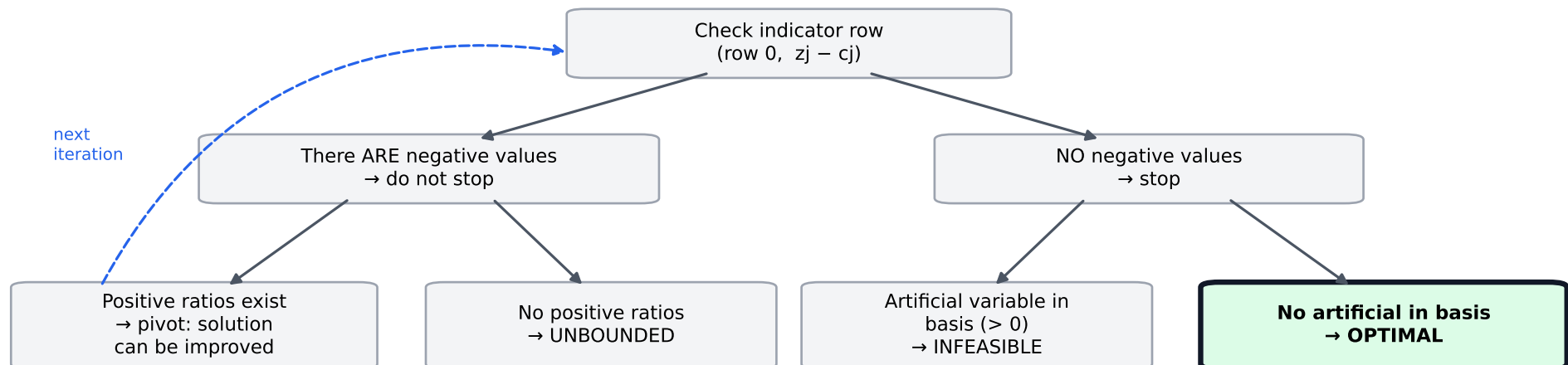
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	0	0	2	0	10
X2	1	0	0.5	1	0.5	0	2.5
X4	2	0	0.5	0	-0.5	1	1.5

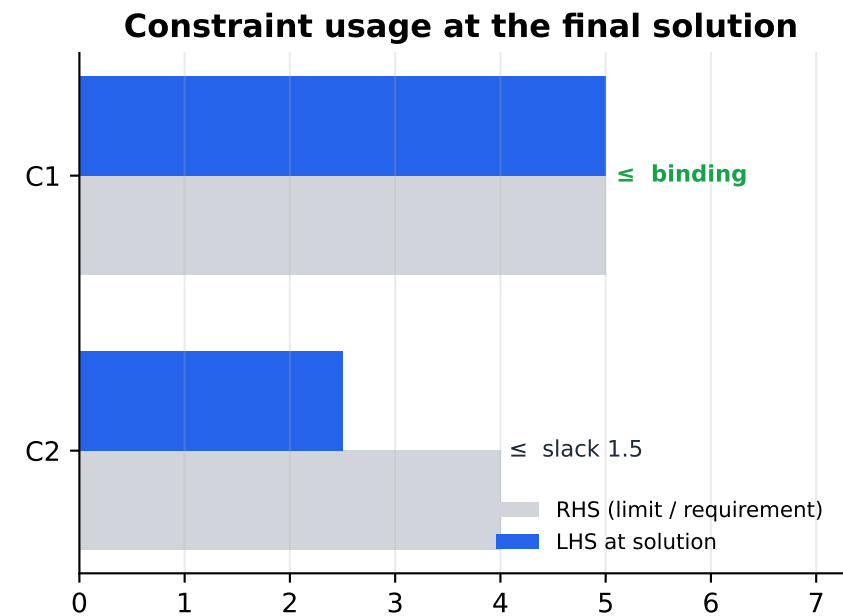
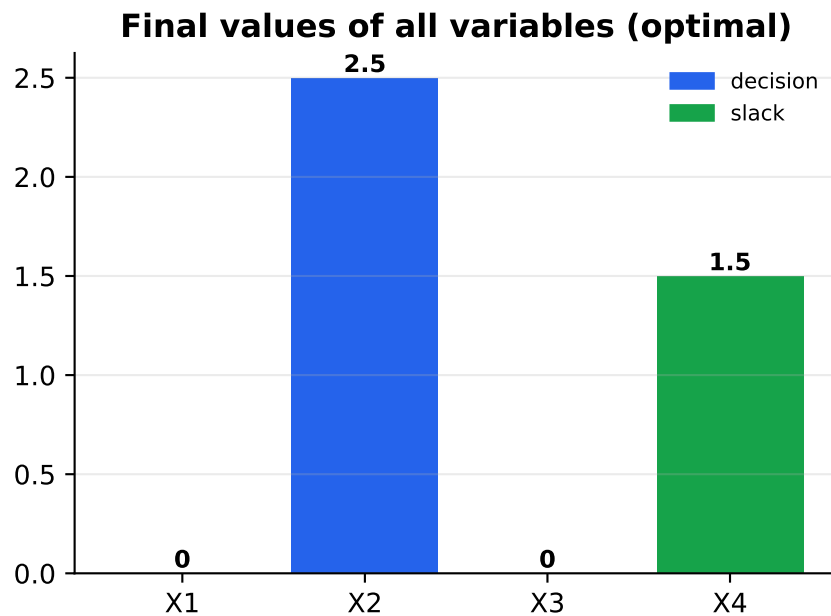
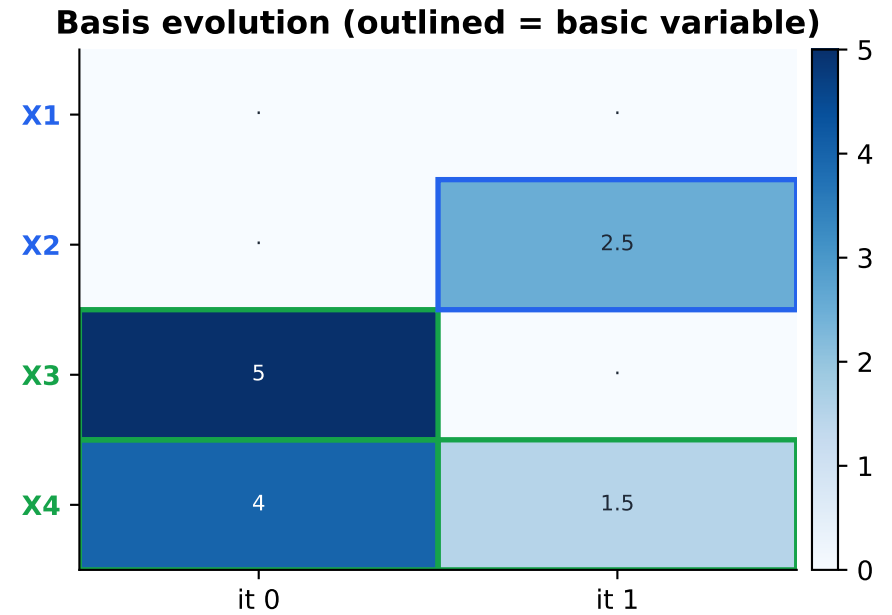
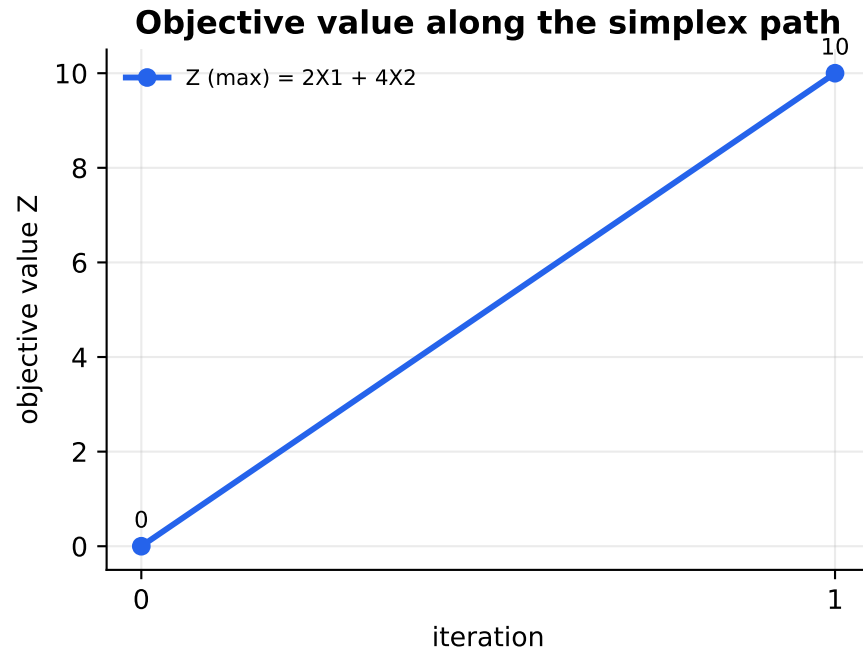
Final check.

- Optimal: row 0 has no negative coefficients. $Z^* = 10$
- Alternative optima: non-basic X1 has a zero coefficient in row 0 - bringing it in changes the solution but not Z.

Simplex decision diagram - outcome: OPTIMAL



Case - Alternative Optimal Solutions - OPTIMAL, $Z^* = 10$



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	10	10
X1	0	0
X2	2.5	2.5

✓ **Results agree**